



FACULTY	Agriculture, Engineering and Natural Sciences		
SCHOOL	Engineering and the Built Environment		
DEPARTMENT	Mechanical and Metallurgical Engineering		
SUBJECT	RIGID BODY DYNAMICS		
SUBJECT CODE	TMED3762		
DATE	November/December 2023		
DURATION	3 Hours	MARKS	100

First Opportunity Examination

Examiner: Prof. I. O. Ohijeagbon

Internal Moderator: Dr. E. Shaanika

External Moderator: Prof. M Waheed (FUNAAB-NIGERIA)

This question paper consists of **3 pages** including this front page.

Instructions

1. Closed book examination
2. Read the questions carefully
3. This question paper consists of **Five** questions
4. Answer **any 4 Questions**
5. Each question carries **25 marks**
6. Neatly drawn sketches should be used where required

Question One

(a) Considering the rectilinear motion of a machine element. Starting from first principles, determine: (i) Velocity as a function of time, (ii) Position as a function of time, and (iii) Velocity as a function of position; of a point on the machine element. **[8 Marks]**

(b) The motion of an automated machine element was investigated by analysing a particle on the element. The particle travels along a straight line with a velocity $v = (12 - 3t^2)m/s$, where t is in seconds. When $t = 1s$, the particle is located 10 m to the left of the origin. Determine the acceleration when $t = 4 s$, the displacement from $t = 0$ to $t = 10 s$, and the distance the particle travels during this time period. **[17 Marks]**

Question Two

The baggage truck A shown in Figure Q. 2 has a weight of 900 N and tows a 550 N cart B and a 325 N cart C. For a short time the driving frictional force developed at the wheels of the truck is $F_A = 40t^3 - 15t^2 - 100t$ N, where t is in seconds.

(a) If the truck starts from rest, (i) obtain expressions for the acceleration and speed in terms of time and (ii) determine its speed after traveling for 3 s. **[11 Marks]**

(b) What is the horizontal force acting on the coupling between the truck and cart B at the instant of 3 s of its journey? Neglect the size of the truck and carts. **[7 Marks]**

(c) Assuming the coupling between carts B and C snaps 5 s after starting from rest. What will be the acceleration of the truck at this instant if the driving force remains the same? **[7 Marks]**

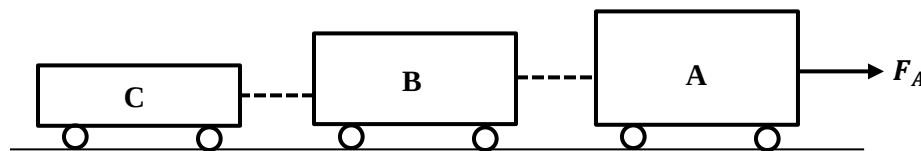


Figure Q. 2

Question Three

(a) The wheel of an industrial machine has an angular acceleration of $\alpha = (0.5\theta) \text{ rad/s}^2$, where θ is in radians. Determine the magnitude of the velocity and acceleration of a point **P** located on its rim after the wheel has rotated 2 revolutions. The wheel has a radius of 0.2 m and assumed to start from rest. **[15 Marks]**

(b) If the wheel in Q. 3 (a) drives a pulley of 1.0 m diameter. Determine the speed of the pulley **Q** in revolution per minute (rpm) and the acceleration at a point **m** located mid-way between the center and outermost part of the pulley **Q**. **[10 Marks]**

Question Four

(a) A planar kinetic analysis of a machine was conducted. And it was required to understand the effect the machine elements has in resisting angular acceleration. Hence, determine the moment of inertia about the axial axis of the cylinder shown in Figure Q. 4 (a). **[5 Marks]**

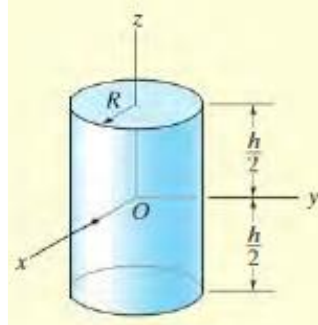
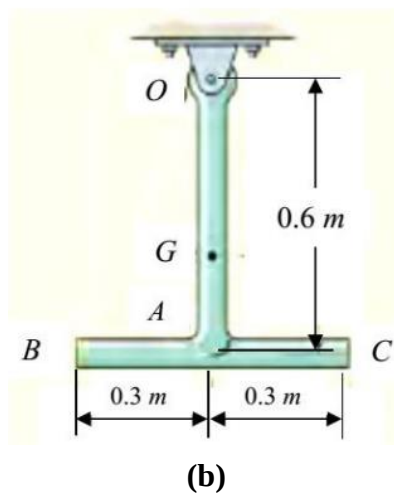
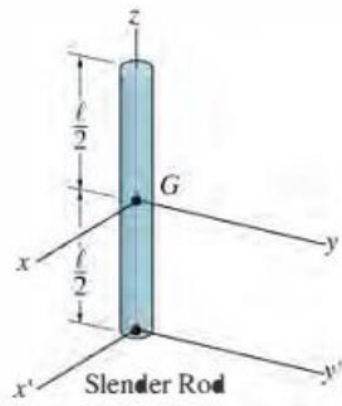


Figure Q. 4 (a)

(b) The pendulum in Figure Q. 4 (b) is suspended from the pin at O and consists of two thin rods, each having a weight of 5 kg . Determine (i) the moment of inertia of the pendulum about an axis passing through point O , (ii) the mass center G of the pendulum relative to point O and (iii) the moment of inertia about G . The moment of inertia of a slender rod shown in Figure Q. 4 (c) is given as: $I_{xx} = I_{yy} = \frac{1}{12} ml^2$; $I_{x'x'} = I_{y'y'} = \frac{1}{3} ml^2$; $I_{z'z'} = 0$ **[20 Marks]**



(b)



(c)

Figure Q. 4

Question Five

(a) A car A has a mass of $2,000 \text{ kg}$ and a center of mass at G . Determine (i) the acceleration, (ii) the reactions at the front and rear wheels, if the rear "driving" wheels are always slipping, whereas the front wheels are free to rotate. Neglect the mass of the wheels. The coefficient of kinetic friction between the wheels and the road is $\mu_k = 0.25$. Take the distance of the front and rear wheels from G to be 1.25 m and 0.75 m , while the height of G is 0.3 m respectively. **[13 Marks]**

(b) Assuming a different car B was designed to function conversely to the car A in Question 5 (a). If the front "driving" wheels of car B are always slipping, whereas the rear wheels are free to rotate. Determine (i) the acceleration, (ii) the reactions at the front and rear wheels, assuming all other specifications of car B are similar to those of car A. **[12 Marks]**